

Evidence Holonomy and Entropy Production: From Universal Coding to Irreversibility

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Abstract

We define an “evidence holonomy” functional on loops of representation transforms applied to sample paths, in two versions. **Representation-space holonomy** scores each representation with its own universal code; by pointwise universality it converges to the entropy-rate difference $h(Q) - h(P)$, where Q is the law of the loop’s output. **KL holonomy** instead scores the original path under the loop’s pushforward law; its rate is the relative-entropy rate $d(P||Q)$, and for the time-reversal loop of a stationary finite-state Markov chain it equals the entropy production rate σ (bits/step). Estimating KL holonomy requires a model of Q that is accurate on paths typical for P , which universality alone does not provide. We analyze the order- R plug-in estimator implemented in our code. It is consistent for σ on Markov chains for every $R \geq 1$. On records observed through a memoryless channel, averages of its order-wise limits are lower bounds on the hidden entropy production. Numerical checks on random chains, closed-form rings, reversible chains, and a hidden ring show convergence, a small positive finite-sample bias, and the gap between observed and hidden irreversibility.

1 Related work

The ingredients are established. For stationary Markov processes, the relative entropy rate between the forward and time-reversed path measures equals the entropy production rate; Gaspard expressed it as the difference between time-reversed and standard entropies per unit time [16]. Roldán and Parrondo proposed estimating this Kullback–Leibler divergence from a single stationary trajectory [13, 17]. They used plug-in estimators on finite blocks and a compression-based estimator built on Ziv–Merhav cross-parsing [18]; the order- R estimator analyzed here is a plug-in estimator of the same kind. Universal estimators based on hitting, return, and waiting times, with consistency and fluctuation results, are due to Chazottes and Redig [19] and Cristadoro et al. [20]. That coarse-grained or partial observation yields lower bounds on dissipation is shown in [12, 13]; waiting-time statistics can reveal irreversibility that transition counts miss [21]. Lossless compression has also been used to quantify order and local entropy production in many-body and active-matter systems [22, 23], and neural networks have been trained to estimate entropy production from trajectories [24].

Our contribution is organizational rather than a new estimator: the loop formalism distinguishes two ways of comparing evidence across a change of representation, identifies which one captures irreversibility, and makes precise why universal coding alone does not justify KL estimation (Section 5). The numerical study documents the behavior of the order- R plug-in estimator, including its finite-sample bias and its behavior on hidden-state records.

2 Setup and Definitions

Alphabet and path space. Fix a finite alphabet $\mathcal{X}$. Let $\mathcal{X}^n$ denote length- n strings and $\mathcal{X}^{\mathbb{N}}$ the one-sided sequence space with its product σ -algebra. Let P be a stationary ergodic probability measure on $\mathcal{X}^{\mathbb{N}}$. Write $X_{0:n-1}$ for the length- n prefix of a sample from P and P_n for its law on $\mathcal{X}^n$.

Universal codes. A *universal code* on alphabet $\mathcal{A}$ is a map $\mathcal{E}_{\mathcal{A}} : \bigcup_{n \geq 1} \mathcal{A}^n \rightarrow \mathbb{R}_+$ assigning a code length in bits to any finite string, such that for every stationary ergodic law Q on $\mathcal{A}^{\mathbb{N}}$,

$$\frac{1}{n} \left(\mathcal{E}_{\mathcal{A}}(Y_{0:n-1}) + \log_2 Q_n(Y_{0:n-1}) \right) \xrightarrow[n \rightarrow \infty]{} 0 \quad Q\text{-a.s. and in } L^1(Q). \quad (1)$$

Classical examples include LZ78, Krichevsky–Trofimov (KT) mixtures over Markov orders, and context-tree weighting [1, 2, 3, 5, 6]. Note that (1) is a statement about paths typical for Q only.

Standing assumptions. Unless stated otherwise, alphabets are finite; processes are stationary and ergodic; and when relative-entropy rates are finite we assume $P_n \ll Q_n$ for all n . All logs are base 2, and all rates are per symbol.

Representation transforms and loops. For each n , let $F_{i,n} : \mathcal{X}_{i-1}^{n_{i-1}(n)} \rightarrow \mathcal{X}_i^{n_i(n)}$ be measurable, where the alphabets $\mathcal{X}_i$ may differ by step and $n_0(n) = n$. A finite list $\gamma = (F_{1,n}, \dots, F_{m,n})$ is a *loop at scale n* if $\mathcal{X}_m = \mathcal{X}_0 = \mathcal{X}$ and $n_m(n) = n + O(1)$. Write $x^{(i)} = F_{i,n} \circ \dots \circ F_{1,n}(x)$ and $L_n = F_{m,n} \circ \dots \circ F_{1,n}$. When $n_m(n) \neq n$, the two arguments being compared are aligned by dropping $O(1)$ boundary symbols, which does not affect per-symbol limits.

Why “holonomy”? In differential geometry, holonomy measures the failure of a vector to return unchanged after parallel transport around a closed loop. Here the “vector” is the description length (or log-likelihood) an observer assigns to a path, and the holonomy is its net change after the path is carried around a loop of representations. Bijective relabelings are “flat” (zero holonomy); time reversal of a nonequilibrium process is not. The analogy is motivational: we use it to organize two different ways of comparing evidence before and after a loop, not to claim a connection or curvature in the geometric sense.

2.1 Two holonomy functionals

Definition 2.1 (Representation-space holonomy). Given universal codes $\mathcal{E}_{\mathcal{X}_i}$ for the intermediate alphabets,

$$\text{Hol}_n^{\text{out}, \gamma}(x) = \sum_{i=1}^m \left(\mathcal{E}_{\mathcal{X}_i}(x^{(i)}) - \mathcal{E}_{\mathcal{X}_{i-1}}(x^{(i-1)}) \right) = \mathcal{E}_{\mathcal{X}}(L_n(x)) - \mathcal{E}_{\mathcal{X}}(x).$$

Definition 2.2 (KL holonomy). Let $Q_n = (L_n) \# P_n$ be the law of the loop’s output. The (ideal) KL holonomy is

$$\text{Hol}_n^{\text{KL}, \gamma}(x) = \log_2 \frac{P_n(x)}{Q_n(x)},$$

the code length of x under Q_n minus its code length under P_n . Its expectation is $\mathbb{E}_P[\text{Hol}_n^{\text{KL}, \gamma}] = D(P_n \| Q_n)$.

Definition 2.2 involves the unknown laws P_n and Q_n ; Section 5 discusses how to estimate it from data.

3 Reductions

Lemma 3.1 (Representation-space holonomy). *Suppose the loop's outputs are the n -marginals of a stationary ergodic process Q (see Section 4.2), and $\mathcal{E}_\mathcal{X}$ is universal. Then*

$$\frac{1}{n} \left(\text{Hol}_n^{\text{out}, \gamma}(X_{0:n-1}) - \log_2 \frac{P_n(X_{0:n-1})}{Q_n(L_n(X_{0:n-1}))} \right) \xrightarrow{n \rightarrow \infty} 0 \quad \text{in } L^1(P). \quad (2)$$

If, in addition, the outputs $L_n(X_{0:n-1})$ are prefixes of a single Q -distributed sequence (as for sliding-block codes), the convergence also holds P -a.s.

Proof. Since $L_n(X_{0:n-1}) \sim Q_n$, the $L^1(Q)$ part of (1) applied to Q controls $\mathcal{E}_\mathcal{X}(L_n(X)) + \log_2 Q_n(L_n(X))$, and applied to P controls $\mathcal{E}_\mathcal{X}(X) + \log_2 P_n(X)$; subtract [4, 5]. L^1 convergence depends only on the marginal laws. Almost-sure convergence needs the outputs for different n to lie on one sample path. This fails for time reversal, where $L_n(x)$ is a different string for every n . $\square$

Theorem 3.2 (Expectation-level identities). *Under the assumptions of Lemma 3.1,*

$$\frac{1}{n} \mathbb{E}_P[\text{Hol}_n^{\text{out}, \gamma}] = h(Q) - h(P) + o(1). \quad (3)$$

For KL holonomy, whenever the limit exists,

$$\frac{1}{n} \mathbb{E}_P[\text{Hol}_n^{\text{KL}, \gamma}] = \frac{1}{n} D(P_n \| Q_n) \xrightarrow{n \rightarrow \infty} d(P \| Q) \geq 0. \quad (4)$$

Proof. Take expectations in (2), using $\mathbb{E}_P[-\log_2 P_n(X)] = H(P_n)$ and $\mathbb{E}_P[-\log_2 Q_n(L_n(X))] = H(Q_n)$, and divide by n . The second identity is Definition 2.2. $\square$

Remark 3.3 (Scope). Equation (3) measures net compression or expansion under the loop and does not depend on which universal code is used. Equation (4) is the *irreversibility* functional: it compares two laws on the same coordinates and is non-negative.

4 Canonical Loops

4.1 Gauge invariance for bijective loops

Corollary 4.1 (Gauge invariance). *If the loop is the identity map on $\mathcal{X}^n$ (for example, a relabeling followed by its inverse), then $Q_n = P_n$, so $\text{Hol}_n^{\text{KL}, \gamma} \equiv 0$ and $\frac{1}{n} \text{Hol}_n^{\text{out}, \gamma}(X_{0:n-1}) = 0$.*

4.2 Coarse-graining and observation channels

A deterministic coarse-graining followed by a lift, $L_n = R_n \circ K_n$, produces outputs supported on the image of R_n . Unless P_n is supported there too, $P_n \not\ll Q_n$ and the KL rate is infinite. So deterministic coarse-grain-lift loops do not give informative KL holonomies. (Estimators with smoothed probabilities return finite numbers that grow as the smoothing vanishes.)

The useful statement about coarse-graining concerns applying the *same* channel to both laws. Let C be a memoryless channel (a stochastic matrix applied independently to each symbol), and write CP for the law of the channel output when the input has law P . By the data-processing inequality [7],

$$D((CP)_n \| (CQ)_n) \leq D(P_n \| Q_n) \quad \text{for every } n. \quad (5)$$

When is the pushforward stationary? For a general sequence of maps L_n , the laws Q_n need not be the n -marginals of any stationary process. A sufficient condition is that the loop acts as a shift-commuting sliding-block code $(L_n(x))_t = F(x_{t-m:t+m})$. Time reversal is not a sliding-block code, but the reversed process of a stationary process is stationary, so it is covered directly.

4.3 Time reversal and entropy production for Markov chains

Let P be a stationary Markov chain on $\mathcal{X} = \{1, \dots, k\}$ with transition matrix T and stationary law π . Its time reversal P^{rev} is the Markov chain with transitions $T_{ij}^* = \pi_j T_{ji} / \pi_i$; equivalently, $P_n^{\text{rev}}(x_0, \dots, x_{n-1}) = P_n(x_{n-1}, \dots, x_0)$.

Time-reversal loop. The loop is literal reversal, $L_n(x_{0:n-1}) = (x_{n-1}, \dots, x_0)$, so $(L_n) \# P_n = P_n^{\text{rev}}$. It can also be written as a loop through the transition alphabet $\mathcal{X}^2$. Encode x as its edge sequence $e_t = (x_{t-1}, x_t)$, $t = 1, \dots, n-1$; reverse the edge sequence; then keep the *second* state of each edge. This yields $(x_{n-1}, x_{n-2}, \dots, x_1)$, the reversal of $x_{1:n-1}$, which differs from literal reversal by one boundary symbol. Our implementation uses this form and compares it against $x_{1:n-1}$. The edge presentation adds no information beyond literal reversal.

Theorem 4.2 (KL holonomy of time reversal equals entropy production). *Suppose $T_{ij} > 0 \Leftrightarrow T_{ji} > 0$ for all i, j . Then*

$$D(P_n \| P_n^{\text{rev}}) = (n-1) \sigma, \quad \sigma = \sum_{i,j} \pi_i T_{ij} \log_2 \frac{\pi_i T_{ij}}{\pi_j T_{ji}} \quad (\text{bits/step}), \quad (6)$$

so $d(P \| P^{\text{rev}}) = \sigma$ and $\frac{1}{n} \mathbb{E}_P[\text{Hol}_n^{\text{KL,rev}}] \rightarrow \sigma$.

Proof. For a path $x = x_{0:n-1}$ with positive probability,

$$\log_2 \frac{P_n(x)}{P_n^{\text{rev}}(x)} = \log_2 \frac{P_n(x_{0:n-1})}{P_n(x_{n-1}, \dots, x_0)} = \log_2 \frac{\pi_{x_0}}{\pi_{x_{n-1}}} + \sum_{t=1}^{n-1} \log_2 \frac{T_{x_{t-1}x_t}}{T_{x_t x_{t-1}}}.$$

Under stationarity, the boundary term has expectation zero and each summand has expectation $\sum_{i,j} \pi_i T_{ij} \log_2(T_{ij}/T_{ji})$. This equals σ because $\sum_{i,j} \pi_i T_{ij} \log_2(\pi_i/\pi_j) = \sum_i \pi_i \log_2 \pi_i - \sum_j \pi_j \log_2 \pi_j = 0$. This is the standard identification of entropy production with the relative entropy between forward and reversed path measures [8, 9, 10, 11]; in this form, for Markov processes, it is due to Gaspard [16]. $\square$

Remark 4.3 (Absolute continuity). If $T_{ij} > 0$ but $T_{ji} = 0$ for some pair, then $P_n \not\ll P_n^{\text{rev}}$ and the rate is $+\infty$.

Remark 4.4 (Why not $h(Q) - h(P)$?). Reversal preserves entropy rates, $h(P) = h(P^{\text{rev}})$, so representation-space holonomy vanishes for the time-reversal loop. Only the KL version detects irreversibility.

General stationary processes. For any stationary P with $P_n \ll P_n^{\text{rev}}$, the same definitions give $\frac{1}{n} \mathbb{E}_P[\text{Hol}_n^{\text{KL,rev}}] \rightarrow d(P \| P^{\text{rev}})$ whenever this limit exists.

5 Estimation

Universality is not enough. A natural idea is to estimate $\log_2 Q_n(x)$ with a universal code trained on samples from Q . But (1) controls code lengths only on paths typical for Q . When $P \neq Q$ are both stationary ergodic, they are mutually singular, and a P -typical path is Q -atypical. So universality says nothing about the code length a Q -trained coder assigns to $X \sim P$. Code-length *differences* from separate compressors (for example, LZ78 on x and on $L_n(x)$) estimate $h(Q) - h(P)$, which is zero for reversal. Estimating $d(P\|Q)$ requires conditional probabilities of Q that are accurate on contexts that occur under P .

The plug-in estimator. Given $x_{0:n-1}$, let $y = x_{1:n-1}$ and $\tilde{y} = (x_{n-1}, \dots, x_1)$ its reversal. For $r = 0, \dots, R$, let $\hat{p}_r(a | c)$ be the KT estimate $(N_y(c, a) + \frac{1}{2}) / (N_y(c) + \frac{k}{2})$ from the counts of context $c \in \mathcal{X}^r$ followed by a in y , and $\hat{q}_r$ the same from $\tilde{y}$. Mix orders with the Bayesian posterior weights obtained by sequentially coding y (respectively $\tilde{y}$), then freeze the resulting predictors $\hat{p}$ and $\hat{q}$. The estimate is

$$\hat{\sigma}_{n,R} = \frac{1}{n-1} \sum_t \log_2 \frac{\hat{p}(y_t | y_{t-R:t-1})}{\hat{q}(y_t | y_{t-R:t-1})}, \quad (7)$$

where the first few symbols use shorter contexts. Both predictors are evaluated on the same y used to fit $\hat{p}$ (an in-sample plug-in estimate of the Kullback–Leibler divergence studied in [13, 17]).

Proposition 5.1 (Consistency for Markov chains). *Let P be a stationary irreducible Markov chain with $T_{ij} > 0 \Leftrightarrow T_{ji} > 0$, and let $R \geq 1$. Then $\hat{\sigma}_{n,R} \rightarrow \sigma$ almost surely.*

Proof sketch. By the ergodic theorem, the empirical frequencies of $(r+1)$ -blocks in y converge a.s. to those of P . Those in $\tilde{y}$ are the same blocks reversed, so they converge to those of P^{rev} . Both P and P^{rev} are first-order Markov. So for every $r \geq 1$ and every context of positive probability, $\hat{p}_r(\cdot | c) \rightarrow T_{c,r,\cdot}$ and $\hat{q}_r(\cdot | c) \rightarrow T_{c,r,\cdot}^*$, and the KT smoothing vanishes. If T is not rank one, the order-0 component’s log-likelihood deficit grows linearly in n , so its posterior weight tends to zero. If T is rank one, the chain is i.i.d., $\sigma = 0$, and every component converges to π . Hence $\hat{p} \rightarrow T$ and $\hat{q} \rightarrow T^*$ uniformly over the finitely many contexts visited. By the support condition, $T_{ij}^* > 0$ whenever $T_{ij} > 0$, so the log-ratios stay bounded. The ergodic theorem applied to (7) gives the limit $\sum_{i,j} \pi_i T_{ij} \log_2(T_{ij}/T_{ij}^*) = \sigma$. $\square$

Remark 5.2 (Robustness to R , not coder independence). Every order $r \geq 1$ is consistent for a first-order chain, and the posterior concentrates on the smallest adequate order. So estimates for different $R \geq 1$ nearly coincide at moderate n ; Section 6 reports agreement to within 10^{-13} bits/step. This reflects a shared, correctly specified model class. It is not evidence that arbitrary universal codes yield the same KL estimate.

Proposition 5.3 (Observed records bound hidden entropy production). *Let Z be a stationary Markov chain satisfying the conditions of Theorem 4.2, with entropy production σ_Z , and let $Y = CZ$ be observed through a memoryless channel C . Write $D_m = D(P_m^Y \| P_m^{Y,\text{rev}})$. Then $D_m \leq (m-1)\sigma_Z$ for every m , so $d(P^Y \| P^{Y,\text{rev}}) \leq \sigma_Z$ whenever the rate exists.*

Proof. Because the channel acts independently on each symbol, reversing the observed path is the same as observing the reversed hidden path: $P_m^{Y,\text{rev}} = (CP^{Z,\text{rev}})_m$. Apply (5) and Theorem 4.2. $\square$

For an observed process Y , Proposition 5.1’s argument shows that the order- R estimator converges (under ergodicity and positivity) to the conditional quantity $d_R = D_{R+1} - D_R$. Because

$D_1 = 0$ (reversal preserves one-symbol marginals), the averages satisfy

$$\frac{1}{m-1} \sum_{R=1}^{m-1} d_R = \frac{D_m}{m-1} \leq \sigma_Z.$$

Individual terms d_R need not be monotone in R or below σ_Z . So a single finite-order estimate is not a certified bound, and the in-sample bias below pushes it upward.

Remark 5.4 (Finite-sample behavior). Evaluating $\hat{p}$ in-sample favors the forward model, so $\hat{\sigma}_{n,R}$ has a positive bias. A heuristic χ^2 argument suggests it is of order $k^{R'+1}/n$, with R' the order the mixture selects; for reversible chains ($\sigma = 0$), Section 6 shows it shrinking roughly in proportion to $1/n$. Fluctuations are of order $n^{-1/2}$. The code provides a block bootstrap for intervals.

6 Numerical validation

All numbers below come from the ancillary files in `anc/. scripts/paper_experiments.py` regenerates them with fixed seeds; the estimator uses $R = 3$ unless stated otherwise.

Random chains. Table 1 and Figure 1 follow 15 random biased chains (five each with 3, 4, and 5 states; σ between 0.0067 and 1.92 bits/step) as the sample length grows from 2^9 to 2^{15} . The median relative error falls from 10.3% at $n = 2^9$ to 1.3% at $n = 2^{15}$. Chains with small σ converge slowest in relative terms; the largest error at $n = 2^{15}$ is 19%.

Table 1: Absolute relative error $|\hat{\sigma} - \sigma|/\sigma$ of the time-reversal estimate ($R = 3$) on 15 random biased chains (five each with 3, 4, and 5 states), by sample length n . Every row uses the same 15 chains.

n	median	interquartile range	max
2^9	10.3%	[2.2%, 14.1%]	123%
2^{11}	8.8%	[5.3%, 11.8%]	86%
2^{13}	4.4%	[3.7%, 7.7%]	31%
2^{15}	1.3%	[0.3%, 3.9%]	19%

Calibration. Figure 2 compares the estimate with the analytic σ for 60 further random chains (3–5 states, bias parameter drawn uniformly from $[0, 0.9]$) at $n = 2^{15}$. The correlation is $r = 0.9994$, and the median relative error is 1.9%. Relative errors are large only where σ itself is tiny: the worst case is a chain with $\sigma \approx 10^{-4}$ bits/step, whose absolute error is at the level of the in-sample bias below. For the 53 chains with $\sigma \geq 0.05$ bits/step, the median relative error is 1.7% and the largest is 10.4%.

Reversible chains and bias. For 10 random detailed-balance chains of each size ($\sigma = 0$), the mean estimate with $R = 3$ is 0.0015 bits/step at $n = 2^{11}$ and 0.00013 at $n = 2^{15}$ for 3-state chains, and 0.0100 and 0.00072 for 5-state chains. Estimates with $R = 1$ and $R = 3$ are identical. The bias shrinks by a factor of 12–14 over a 16-fold increase in n , consistent with the roughly $1/n$ behavior described in Section 5.

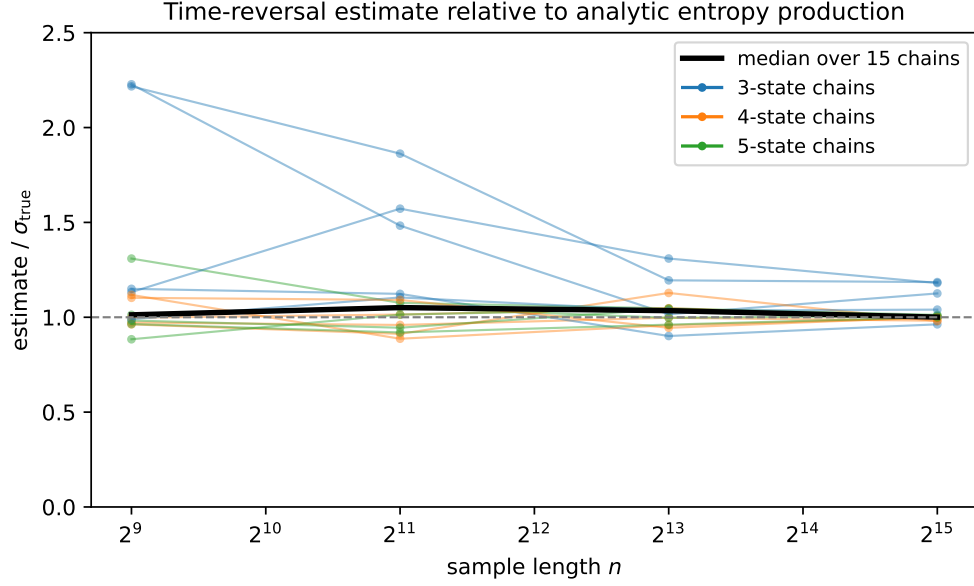


Figure 1: Estimate divided by the analytic σ for 15 random chains (thin lines, colored by state count) and their median (black), as the sample length grows.

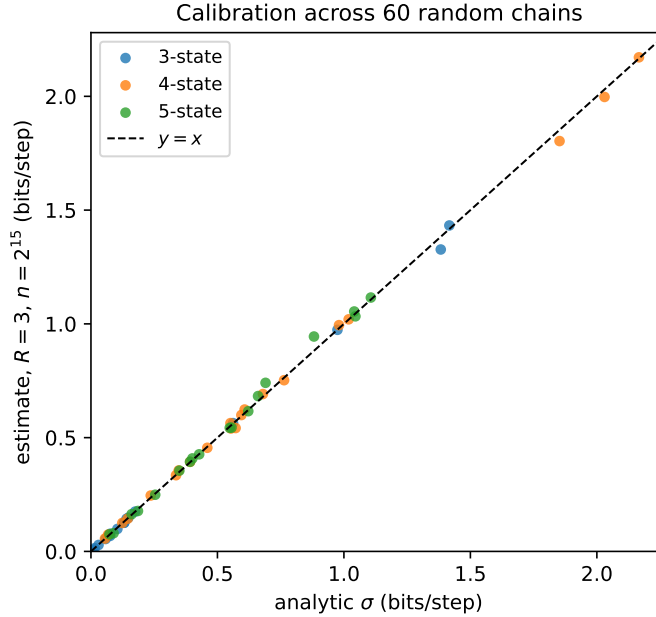


Figure 2: Estimate ($R = 3$, $n = 2^{15}$) against analytic entropy production for 60 random chains.

Order robustness and closed forms. On first-order chains, estimates with $R = 1, 2, 3, 5$ agree to within 10^{-13} bits/step, as the remark in Section 5 predicts. The validation battery (`uec-battery --run_suite`) also checks:

- three-state rings with closed-form σ , to within 5% relative error at $n = 1.5 \times 10^5$;
- a reversible two-state chain (estimate below 5×10^{-4});

- gauge loops (below 5×10^{-3});
- agreement with a smoothed transition-count estimator;
- block-bootstrap coverage of σ ;
- negative controls: alphabet mismatch, empty outputs, and self-evidence differences, which stay near zero under reversal even when $\sigma > 0$.

Hidden-state records. Figure 3 observes a three-state ring ($p = 0.3$ forward, $q = 0.05$ backward, $\sigma_Z = 0.646$ bits/step) through two channels, for $n = 2^{17}$ and three seeds. Through a noisy channel that reports the true state with probability 0.8, the estimates for $R = 1, \dots, 6$ range from 0.085 to 0.127 bits/step, well below σ_Z . They rise from $R = 1$ to $R = 3$ and then stay the same: at this sample size the mixture puts negligible weight on orders above 3, so larger R adds nothing. Through a channel that merges states 1 and 2, the observation is symmetric under the relabeling that maps the ring to its reversal. The observed process is therefore exactly reversible, and the estimates stay at most 7×10^{-5} bits/step even though $\sigma_Z > 0$.

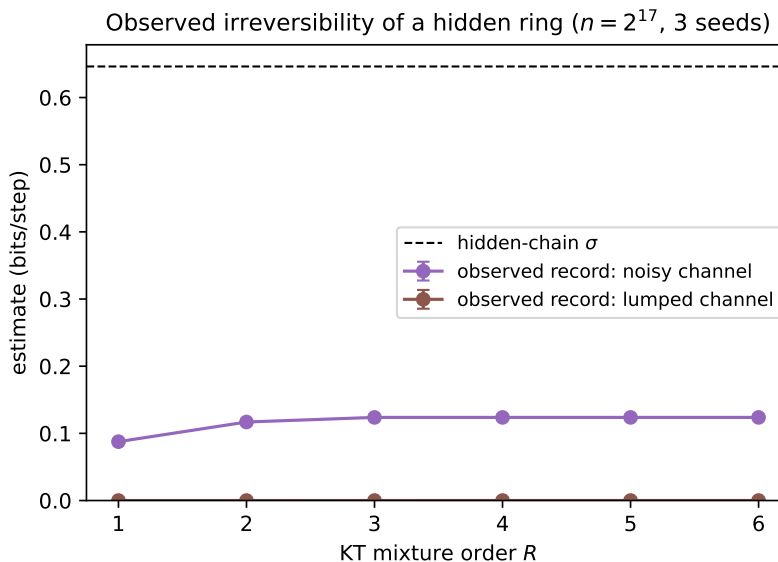


Figure 3: Estimates on observed records of a hidden three-state ring, by KT mixture order R (mean and range over three seeds), against the hidden chain’s entropy production.

Arrow-of-time demonstrations. The package also applies the estimator to quantized real-valued signals (audio, sensor, and price series; images and video via vector quantization). It compares held-out windows with their literal reversals and reports an AUC with a block-bootstrap interval. On i.i.d. noise with 10^4 samples, single-run AUCs ranged from 0.11 to 0.74 across seeds. So these demonstrations are illustrations whose intervals must be read, not validation of the theory.

7 Code and data

Code, ancillary data, and scripts are at <https://github.com/jbwinters/Evidence-Holonomy>. `scripts/reproduce_all.sh` regenerates `anc/`, the table, the figures, and this document.

Discussion

The two holonomies answer different questions. Evaluating each representation with its own universal code gives the entropy-rate difference $h(Q) - h(P)$: it is insensitive to the choice of code and detects net compression or expansion, but it vanishes for time reversal. Evaluating the original path under the loop’s pushforward law gives the relative-entropy rate $d(P\|Q)$, which for Markov time reversal is the entropy production rate. The second quantity is the one that captures irreversibility. It requires a consistent model of the reversed process on forward-typical contexts, and a finite-order plug-in estimator supplies one for finite-state Markov chains.

Extensions. Quantizing a continuous process and observing a Markov chain through a memoryless channel are both instances of (5). The observed irreversibility rate therefore bounds the underlying entropy production from below [12, 13]; estimators that use waiting times can recover more of it than finite-order transition statistics [21]; the quantum analogue follows from the monotonicity of quantum relative entropy [14, 15]. Turning finite-sample, finite-order estimates into certified bounds would require controlling both the in-sample bias and the order-wise terms d_R .

Limitations. The results assume finite alphabets, stationarity, and ergodicity. The consistency result covers finite-state Markov chains; for hidden-state records, the estimator targets the conditional quantities d_R , not σ_Z . Estimates carry a positive finite-sample bias (empirically of order $1/n$) and $O(n^{-1/2})$ fluctuations. The loop formalism organizes these estimators but does not by itself supply new estimators for loops other than time reversal.

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References

- [1] J. Ziv and A. Lempel, “Compression of individual sequences via variable-rate coding,” *IEEE Trans. Inf. Theory*, 24(5):530–536, 1978.
- [2] R. E. Krichevsky and V. K. Trofimov, “The performance of universal encoding,” *IEEE Trans. Inf. Theory*, 27(2):199–207, 1981.
- [3] F. M. J. Willems, Y. M. Shtarkov, and T. J. Tjalkens, “The Context-Tree Weighting Method: Basic Properties,” *IEEE Trans. Inf. Theory*, 41(3):653–664, 1995.
- [4] A. R. Barron, “The strong ergodic theorem for densities: generalized Shannon–McMillan–Breiman theorem,” *Annals of Probability*, 13(4):1292–1303, 1985.
- [5] P. C. Shields, *The Ergodic Theory of Discrete Sample Paths*, Graduate Studies in Mathematics, Vol. 13, American Mathematical Society, 1996.
- [6] I. Csiszár and P. C. Shields, “Information theory and statistics: A tutorial,” *Foundations and Trends in Communications and Information Theory*, 1(4):417–528, 2004.

- [7] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.
- [8] J. Schnakenberg, “Network theory of microscopic and macroscopic behavior of master equation systems,” *Reviews of Modern Physics*, 48(4):571–585, 1976.
- [9] G. E. Crooks, “Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences,” *Phys. Rev. E*, 60:2721–2726, 1999.
- [10] R. Kawai, J. M. R. Parrondo, and C. Van den Broeck, “Dissipation: The phase-space perspective,” *Phys. Rev. Lett.*, 98:080602, 2007.
- [11] U. Seifert, “Stochastic thermodynamics, fluctuation theorems and molecular machines,” *Reports on Progress in Physics*, 75:126001, 2012.
- [12] A. Gomez-Marin, J. M. R. Parrondo, and C. Van den Broeck, “Lower bounds on dissipation upon coarse graining,” *Phys. Rev. E*, 78:011107, 2008.
- [13] É. Roldán and J. M. R. Parrondo, “Estimating Dissipation from Single Stationary Trajectories,” *Phys. Rev. Lett.*, 105:150607, 2010.
- [14] G. Lindblad, “Completely positive maps and entropy inequalities,” *Communications in Mathematical Physics*, 40:147–151, 1975.
- [15] A. Uhlmann, “Relative entropy and the Wigner–Yanase–Dyson–Lieb concavity in an interpolation theory,” *Communications in Mathematical Physics*, 54:21–32, 1977.
- [16] P. Gaspard, “Time-reversed dynamical entropy and irreversibility in Markovian random processes,” *J. Stat. Phys.*, 117:599–615, 2004.
- [17] É. Roldán and J. M. R. Parrondo, “Entropy production and Kullback–Leibler divergence between stationary trajectories of discrete systems,” *Phys. Rev. E*, 85:031129, 2012.
- [18] J. Ziv and N. Merhav, “A measure of relative entropy between individual sequences with application to universal classification,” *IEEE Trans. Inf. Theory*, 39(4):1270–1279, 1993.
- [19] J.-R. Chazottes and F. Redig, “Testing the irreversibility of a Gibbsian process via hitting and return times,” *Nonlinearity*, 18:2477–2489, 2005.
- [20] G. Cristadoro, M. Degli Esposti, V. Jakšić, and R. Raquépas, “Recurrence times, waiting times and universal entropy production estimators,” *Lett. Math. Phys.*, 2023; arXiv:2209.09716.
- [21] I. A. Martínez, G. Bisker, J. M. Horowitz, and J. M. R. Parrondo, “Inferring broken detailed balance in the absence of observable currents,” *Nat. Commun.*, 10:3542, 2019.
- [22] S. Martiniani, P. M. Chaikin, and D. Levine, “Quantifying hidden order out of equilibrium,” *Phys. Rev. X*, 9:011031, 2019.
- [23] S. Ro, B. Guo, A. Shih, T. V. Phan, R. H. Austin, D. Levine, P. M. Chaikin, and S. Martiniani, “Model-free measurement of local entropy production and extractable work in active matter,” *Phys. Rev. Lett.*, 129:220601, 2022.
- [24] D.-K. Kim, Y. Bae, S. Lee, and H. Jeong, “Learning entropy production via neural networks,” *Phys. Rev. Lett.*, 125:140604, 2020.